Recap: Lecture 28

In Lecture 28, we

- Introduced the Fourier transform and inverse FT
- Connected the FT with Fourier series
- Gave a theorem for existence of the FT

$$\mathscr{F}\{f\} = \int_{-\infty}^{\infty} f(x)\, \mathrm{e}^{-\mathrm{i}\,k\,x}\, \mathrm{d}x = \widehat{f}(k)$$

$$\mathscr{F}^{-1}\{f\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(k)\, \mathrm{e}^{+\mathrm{i}\,k\,x}\, \mathrm{d}k$$

Lecture 29

In Lecture 29, we will:

1. Introduce a FT table for simpler evaluation of FTs and inverse FTs
2. Introduce convolution and shift theorems for FT
3. Learn rules to help apply FTs to ODEs

Table of Fourier transforms

$f(x)$	$\widehat{f}(k)$
$\delta_a(x) \quad (a \in \mathbb{R})$	$\mathrm{e}^{-\mathrm{i}ak}$
$\mathrm{e}^{-a\lvert x\rvert} \quad (a > 0)$	$\dfrac{2a}{a^2 + k^2}$
$H_{-a}(x) - H_a(x) \quad (a > 0)$	$2a \operatorname{sinc} ak$
$\mathrm{e}^{-ax^2} \quad (a > 0)$	$\sqrt{\dfrac{\pi}{a}}\mathrm{e}^{-\frac{k^2}{4a}}$

$f(x)$	$\widehat{f}(k)$
$\mathrm{e}^{\mathrm{i}ax} \quad (a \in \mathbb{R})$	$2\pi\delta_a(k)$
$\dfrac{1}{a^2 + x^2} \quad (a > 0)$	$\dfrac{\pi}{a}\mathrm{e}^{-a\lvert k\rvert}$
$\operatorname{sinc}(ax) \quad (a > 0)$	$\dfrac{\pi}{a}\left(H_{-a}(k) - H_a(k)\right)$
$\operatorname{sech}(ax) \quad (a \neq 0)$	$\dfrac{\pi}{a}\operatorname{sech}\left(\dfrac{\pi k}{2a}\right)$

$$^*\operatorname{sinc}(y) = \begin{cases} \frac{\sin y}{y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Exercise 6.4: Find the inverse Fourier transform of $\widehat{f}(k) = \sin(2k)$

We write the sine function in terms of complex exponentials and find

$$\begin{aligned}\hat{f}(k) &= \frac{1}{2i}\Big(e^{2ik} - e^{-2ik}\Big) \\ &= \frac{1}{2i}\mathscr{F}\Big\{\delta(x+2) - \delta(x-2)\Big\} \qquad \text{(using table)}\end{aligned}$$

Hence

$$\begin{aligned}\mathscr{F}^{-1}\{\widehat{f}(k)\} &= \mathscr{F}^{-1}\Big\{-\frac{i}{2}\mathscr{F}\{\delta(x+2) - \delta(x-2)\}\Big\} \\ &= -\frac{i}{2}\mathscr{F}^{-1}\Big\{\mathscr{F}\big\{\delta(x+2) - \delta(x-2)\big\}\Big\} \qquad \text{(linearity)} \\ &= \frac{i}{2}\Big(\delta(x-2) - \delta(x+2)\Big)\end{aligned}$$

Recap: Convolutions for Laplace transforms

- Given two functions, f and g, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$(f * g)(t) = \int_0^t f(t - \tau) g(\tau) \, \mathrm{d}\tau$$

- Laplace transforms convert convolutions into multiplication, i.e.,

$$\mathscr{L}\{f * g\} = F(s)G(s)$$

Convolutions for Fourier transforms

- We define the convolution of $f(x)$ with $g(x)$ for $x \in \mathbb{R}$ as

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - \xi) g(\xi) \, \mathrm{d}\xi$$

- Then the Fourier transform of the convolution is

$$\mathscr{F}\{f * g\} = \widehat{f}(k) \, \widehat{g}(k)$$

Exercise 6.5: Calculate the convolution of

$$f(x) = \frac{1}{x^2+4} \quad \text{with} \quad g(x) = \frac{1}{x^2+9}$$

We compute the convolution using its Fourier transform.

$$\mathscr{F}\{f * g\} = \hat{f}(k)\hat{g}(k) = \left[\frac{\pi}{2}e^{-2|k|}\right]\left[\frac{\pi}{3}e^{-3|k|}\right] = \frac{5\pi}{6}\,\frac{\pi}{5}e^{-5|k|} = \frac{5\pi}{6}\mathscr{F}\{\frac{1}{x^2+25}\}$$

Hence

$$f * g(x) = \frac{5\pi}{6}\,\frac{1}{x^2+25}$$

Recap: Laplace transform shift theorems

Theorem (t-shift)

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{ f(t-c)H(t-c) \Big\} = \mathrm{e}^{-cs} F(s), \quad \mathrm{Re}(s) > \alpha$$

Theorem (s-shift)

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega t} f(t)\} = F(s-\omega), \qquad \mathrm{Re}(s) > \alpha + \omega$$

Fourier transform shift theorems

1. $\mathscr{F}\{f(x-y)\} = \mathrm{e}^{-iky}\,\widehat{f}(k) \qquad (y \in \mathbb{R})$
2. $\mathscr{F}\{\mathrm{e}^{iqx} f(x)\} = \widehat{f}(k-q) \qquad (q \in \mathbb{R})$

Exercise 6.6: Evaluate the following

$$\mathscr{F}\{\mathrm{e}^{-a|x-2|}\} = \frac{2a\mathrm{e}^{-2ik}}{a^2+k^2}$$

$$\mathscr{F}^{-1}\left\{\operatorname{sech}\frac{\pi(k-5)}{2}\right\} = \frac{1}{\pi}\,\mathrm{e}^{5ix}\operatorname{sech} x$$

Application to ODEs

Just like Laplace transforms, Fourier transforms turns calculus into algebra!

1. Differentiation wrt x turns into multiplication by k

$$\mathscr{F}\left(f^{(n)}\right) = (ik)^n \widehat{f}$$

We will primarily use $\mathscr{F}(f') = ik\widehat{f}$ *and* $\mathscr{F}(f'') = -k^2\widehat{f}$

2. Multiplication by x turns into differentiation w.r.t. k

$$\mathscr{F}(x^n f) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

These identities assume that the function $f(x)$ and its derivatives decay sufficiently rapidly at $\pm\infty$.

Exercise 6.7: Show that $\mathscr{F}\{f'(x)\} = ik\mathscr{F}\{f(x)\}$

Using integration by parts we find

$$\begin{aligned}\mathscr{F}\{f'(x)\} &= \int_{-\infty}^{\infty} f'(x)e^{-ikx}\,dx \\ &= \left[f(x)e^{-ikx}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x)(-ik)e^{-ikx}\,dx \\ &= ik\hat{f}(k)\end{aligned}$$

This follows as the boundary term vanishes due to the rapid decay of $f(x)$ at $x \to \pm\infty$.

Exercise 6.8: Show that $\mathscr{F}\{x\, f(x)\} = i\widehat{\frac{\mathrm{d}f}{\mathrm{d}k}}$

$$\begin{aligned}
\mathscr{F}\{xf(x)\} &= \int_{-\infty}^{\infty} f(x) x e^{-ikx}\, dx \\
&= \int_{-\infty}^{\infty} f(x)\, i \frac{d}{dk} e^{-ikx}\, dx \\
&= i\frac{d}{dk}\hat{f}(k)
\end{aligned}$$

Exercise 6.9: Find the solution to the ODE

$$y''(x) - y(x) = \delta(x)$$

subject to the condition that the Fourier transform of $y(x)$ exists.

We apply the Fourier transform to both sides and obtain

$$-k^2\hat{y} - \hat{y} = 1$$

and hence

$$\hat{y}(k) = -\frac{1}{1+k^2} = -\frac{1}{2}\mathscr{F}\{e^{-|x|}\}$$

The final solution is thus

$$y(x) = -\frac{1}{2}\, e^{-|x|}$$